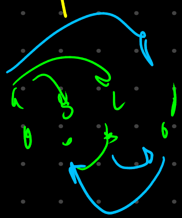
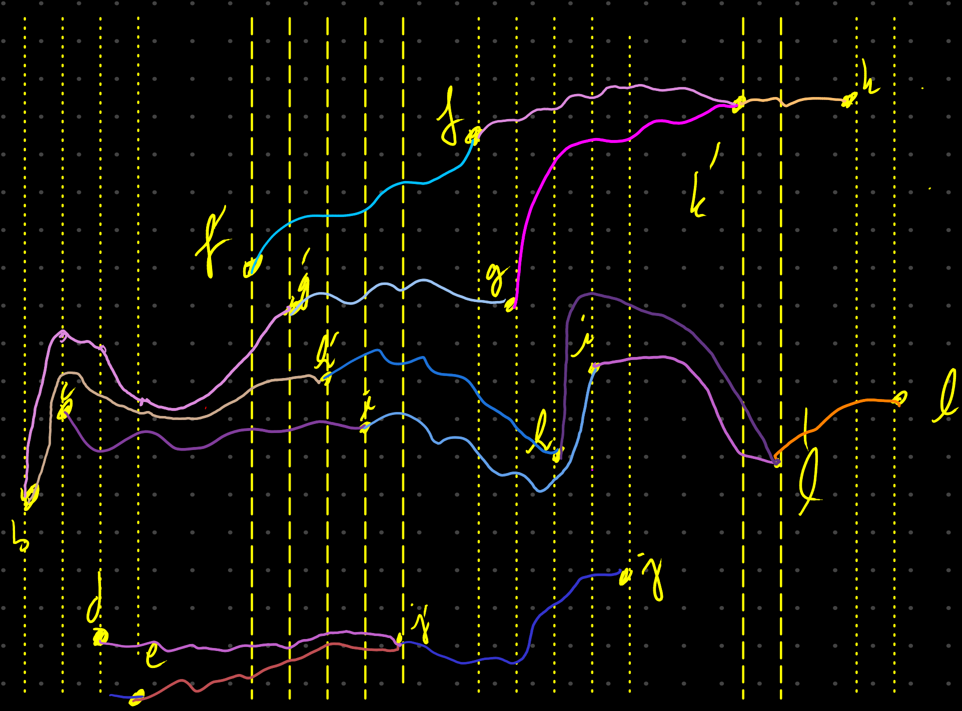
Let's add one edge to the counter-example graph

(a, d)





$\varphi(ab)=1$ $\varphi(ac)=x$ $\varphi(ad)=1$ $\varphi(ae)=1$

$\varphi(bc)=x$ $\varphi(bd)=0$ $\varphi(be)=0$

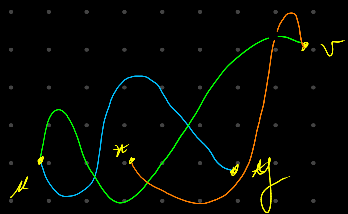
$\varphi(cd)=1$ $\varphi(ce)=1$ $\varphi(de)=x$

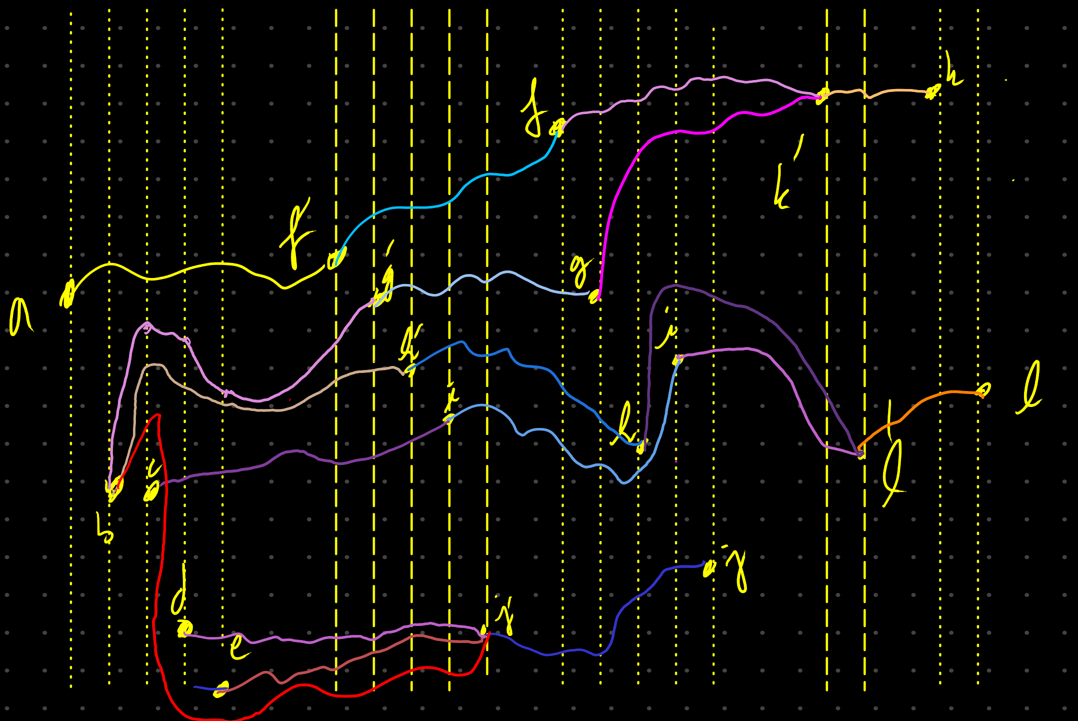
$\varphi(fg)=\cancel{1}$ $\varphi(fh)=1$ $\varphi(fi)=\cancel{1}$ $\varphi(fj)=1$

$\varphi(gh)=1$ $\varphi(gi)=1$ $\varphi(gj)=0$ $\varphi(hi)=1$

$\varphi(h\;)=0$ $\varphi(ij)=1$

what if we actually have a cycle





$\varphi(cd)=1$

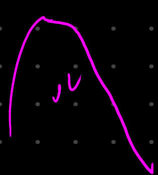
$\varphi(ce)=1$

$\varphi(de)=1$

$\varphi(bd)=0$

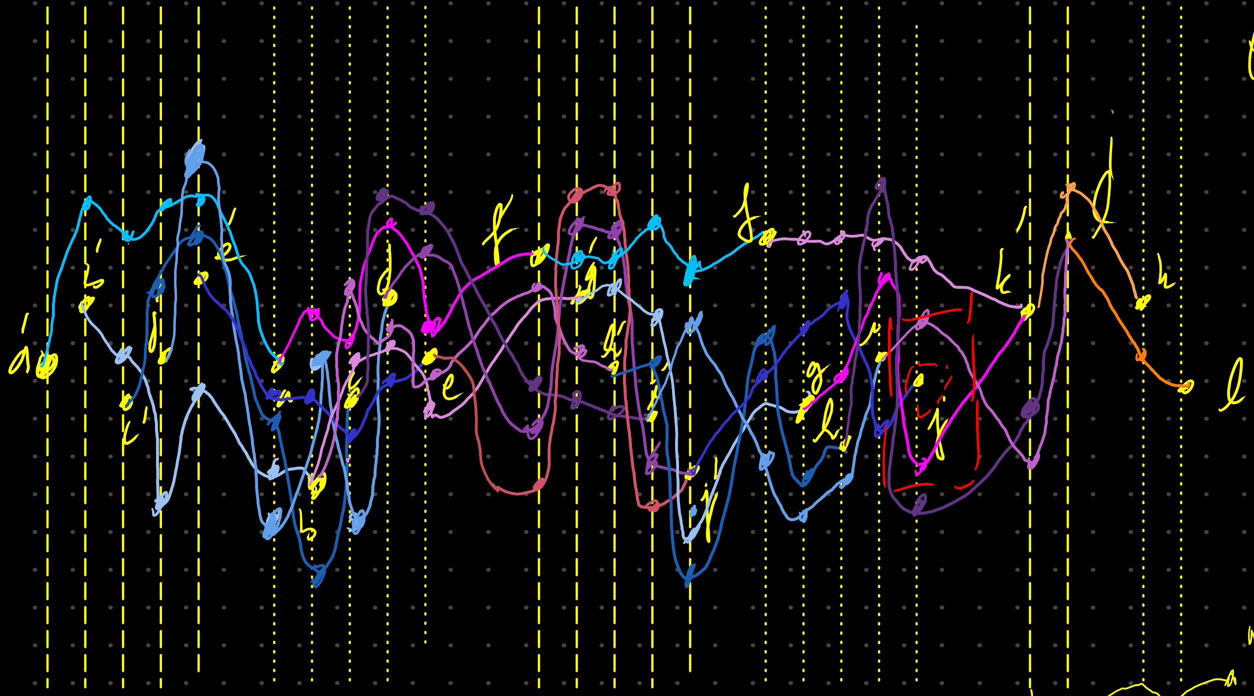
$\varphi(be)=1$

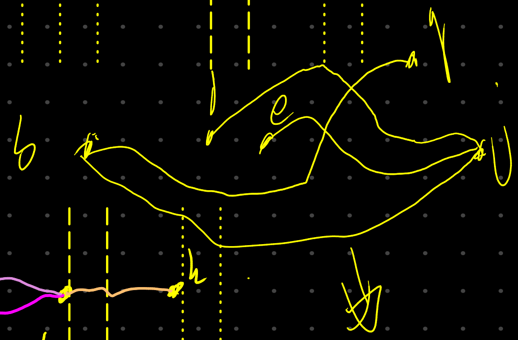
$\varphi(bc)=$

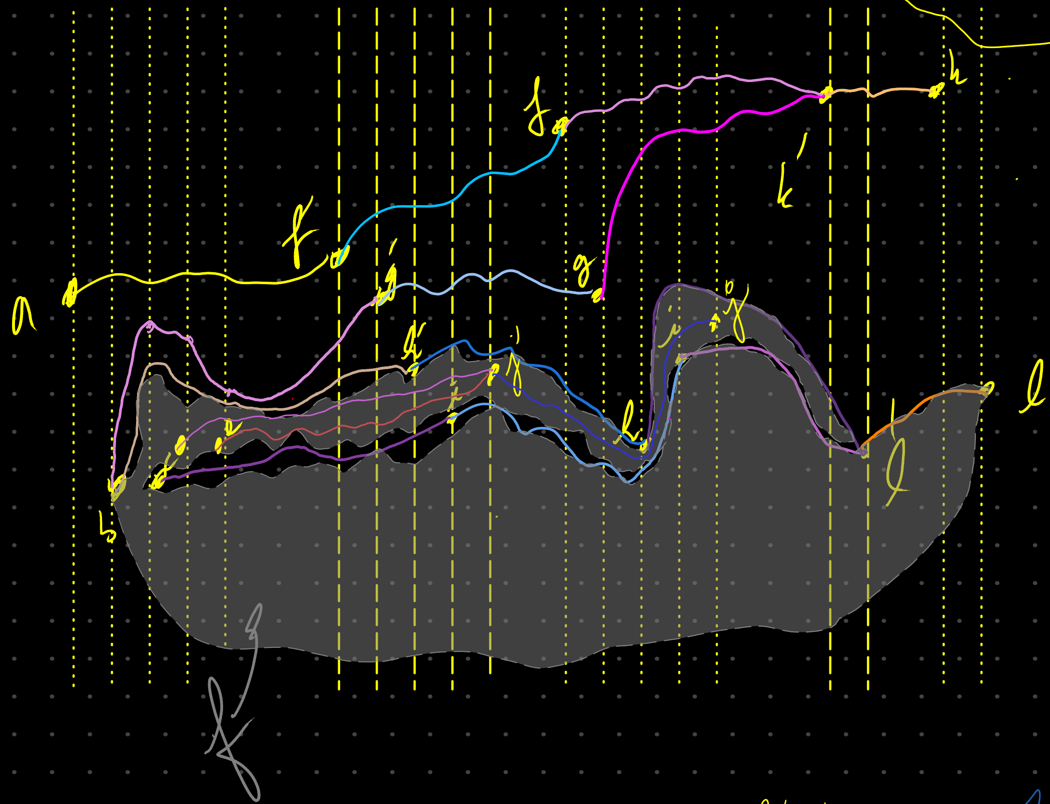


$\varphi(k$

$\bar{\bar{e}}|$







$\varphi(ab)=1$ $\varphi(ac)=x$ $\varphi(ad)=1$ $\varphi(ae)=1$

$\varphi(bc)=x$ $\varphi(bd)=0$ $\varphi(be)=0$

$\varphi(cd)=1$ $\varphi(ce)=1$ $\varphi(de)=x$

$\varphi(fg)=\cancel{1}$ $\varphi(fh)=1$ $\varphi(fi)=1$ $\varphi(fj)=1$

$\varphi(gh)=1$ $\varphi(gi)=1$ $\varphi(gj)=0$ $\varphi(hi)=1$

$\varphi(hj)=0$ $\varphi(ij)=1$

$\boxed{\varphi(db)=1\to 0}$ $\varphi(ea)=0$

$\varphi(da)=0$ $\boxed{\varphi(eb)=1\to 0}$

$\varphi(dc)=0$ $\varphi(ec)=0$

$\varphi(de)=1$ $\varphi(ef)=0$

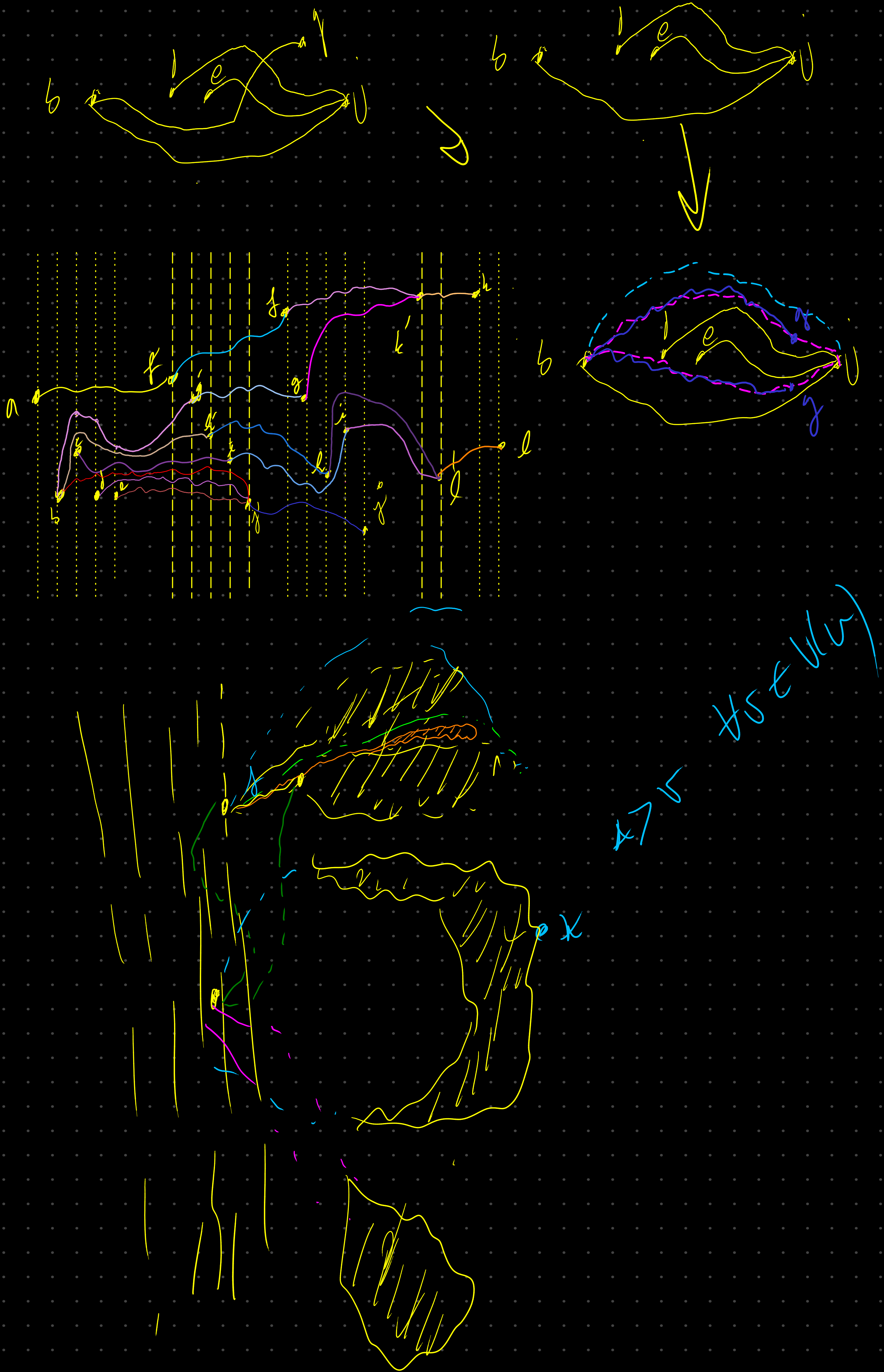
$\varphi($

$\varphi(ji)=0$ $\varphi(jh)=1$

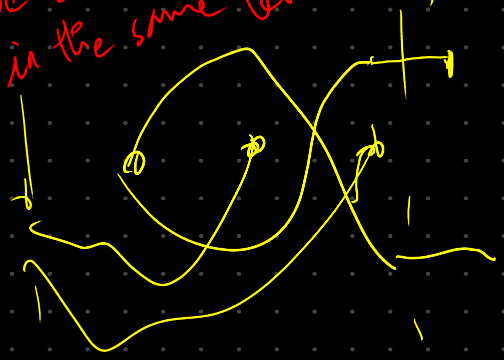
y

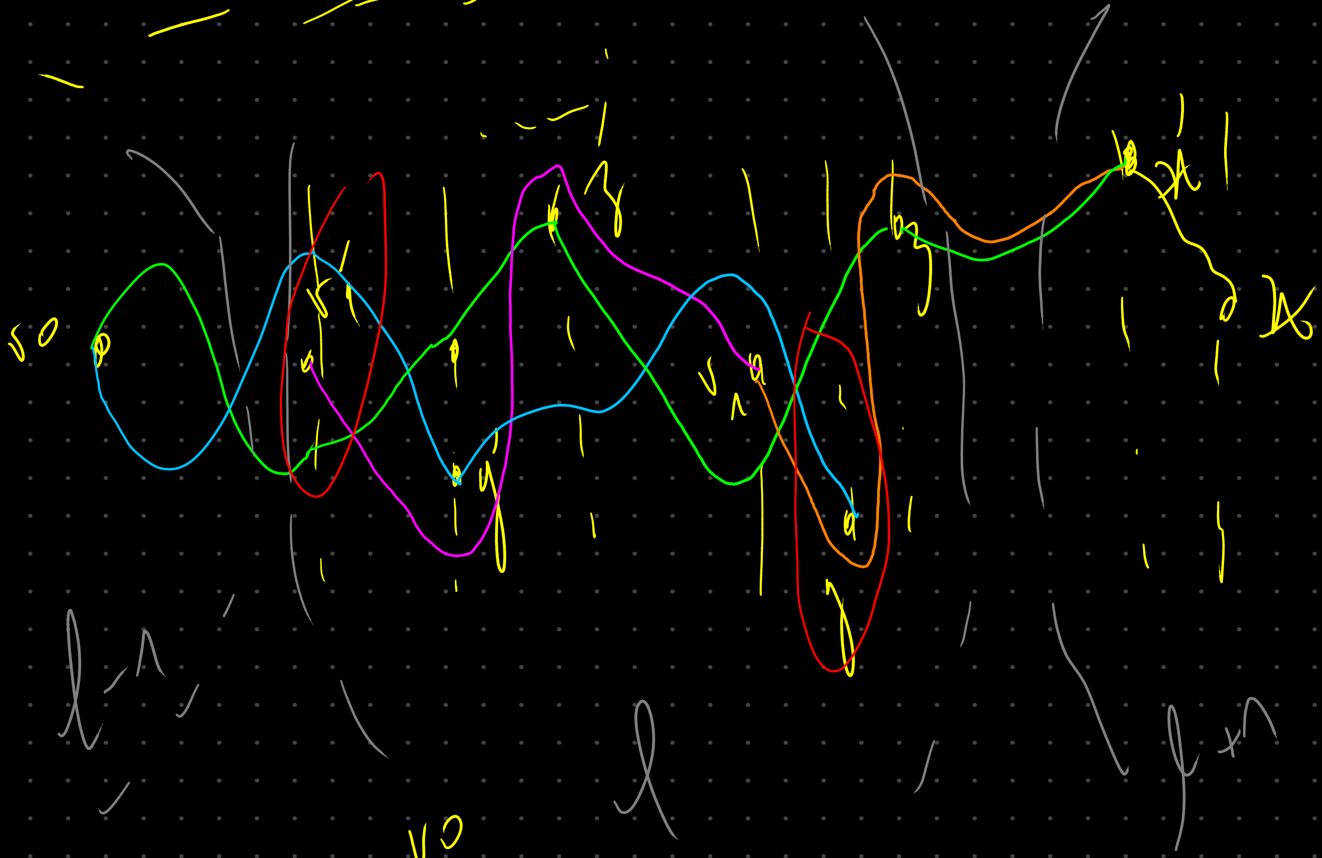
0

$\varphi(hj)=0$ $\varphi(ij)=1$



we can't have this in the same level.





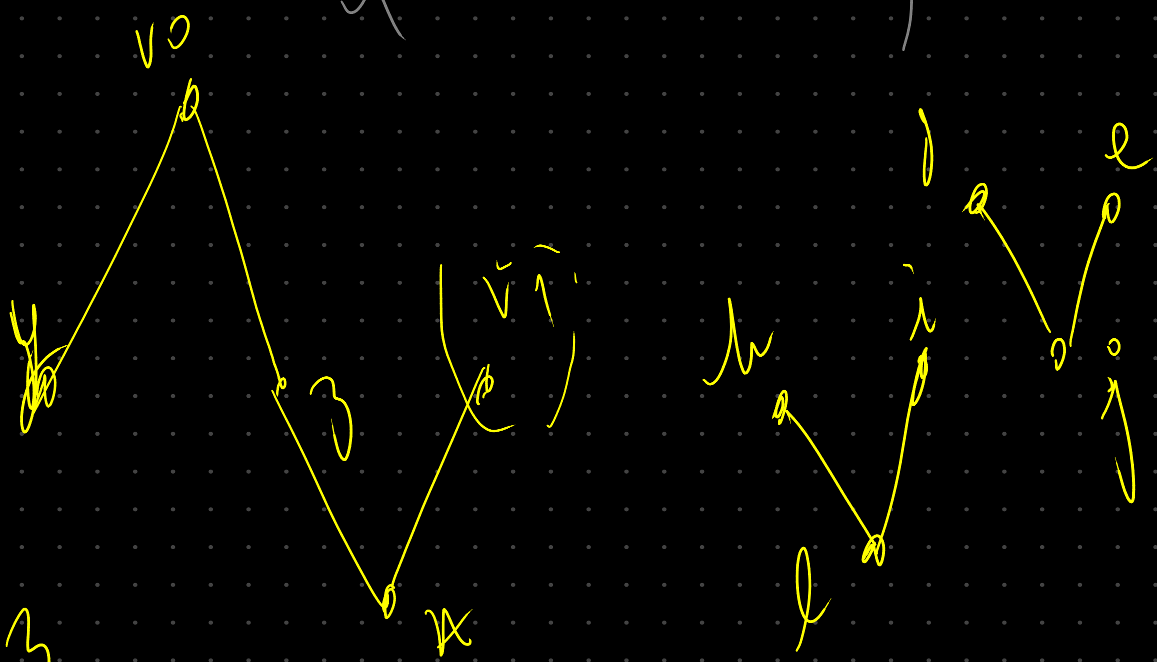
i<j

i<y

y<i

y<z

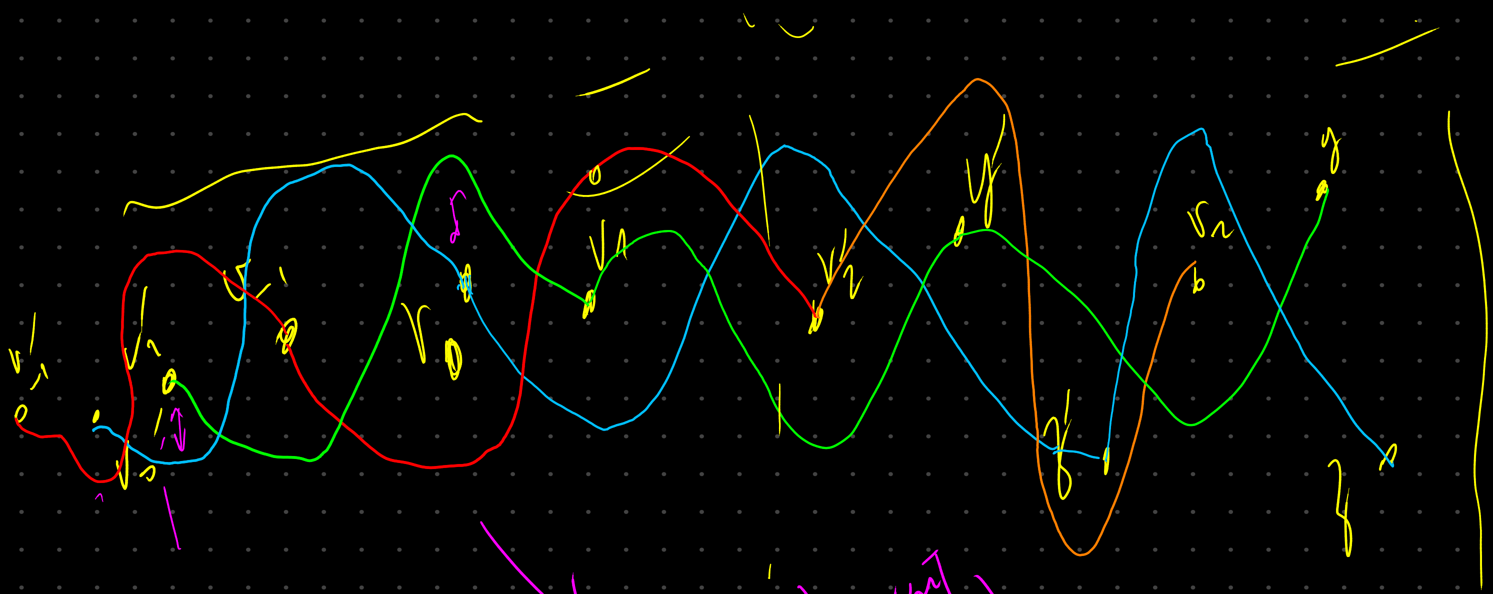
y<v1<z



I think I found the relation

$y < 3$

$3 < v_2 < 4$

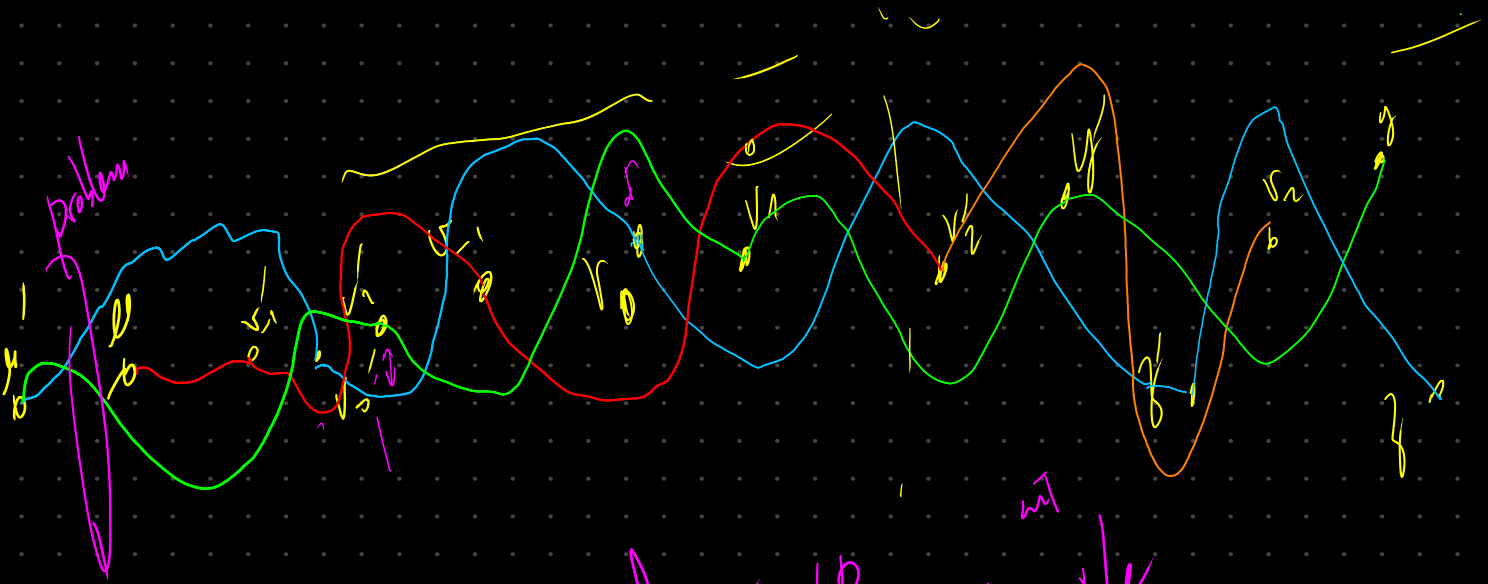


$\varphi(v_{-1}, v_1) = 1$

$\varphi(v_{-1}, v_0) = 0$

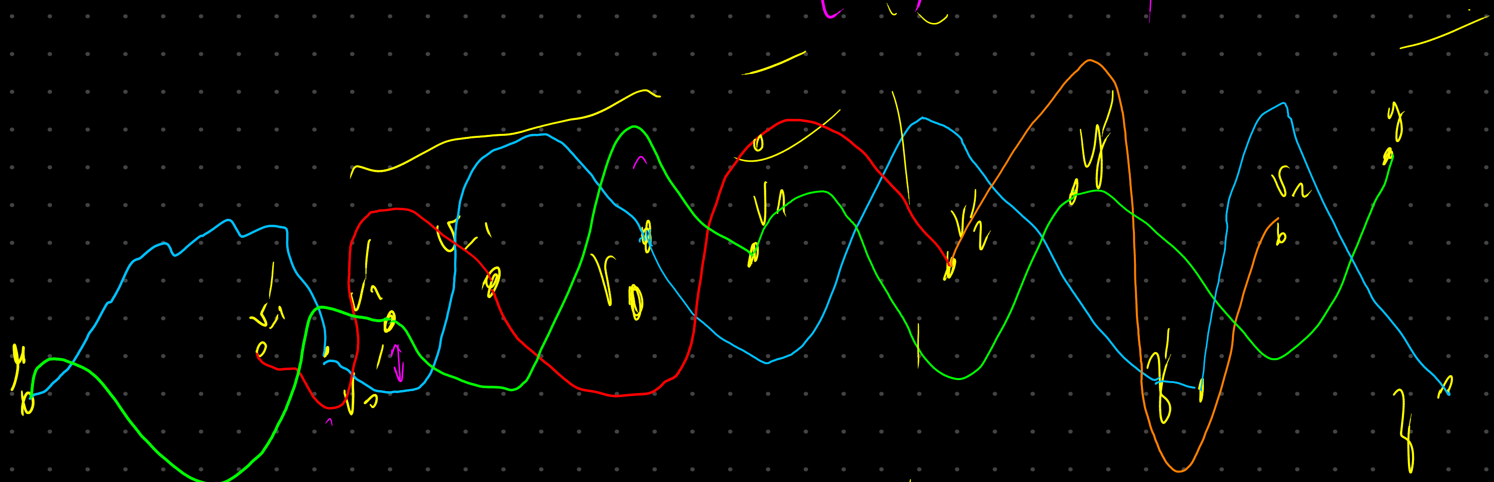
$\varphi(v_1, v_2) = 0$

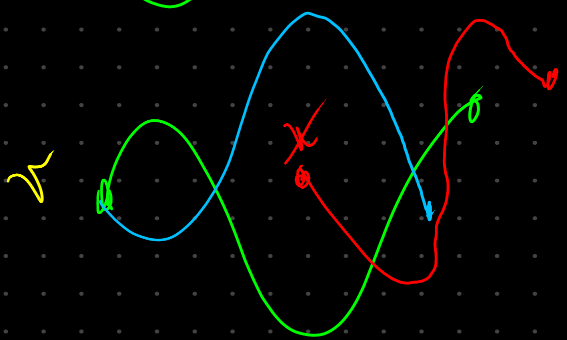
it's not relevant if the connected lamps don't come from one at the end, (we can draw them independently)



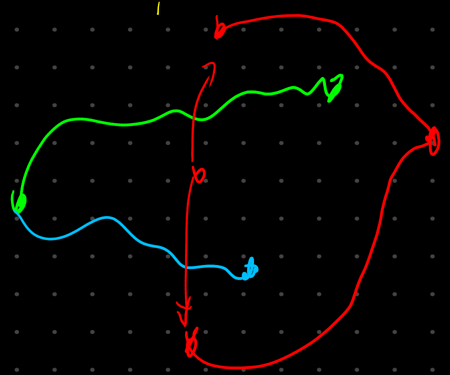
⇓

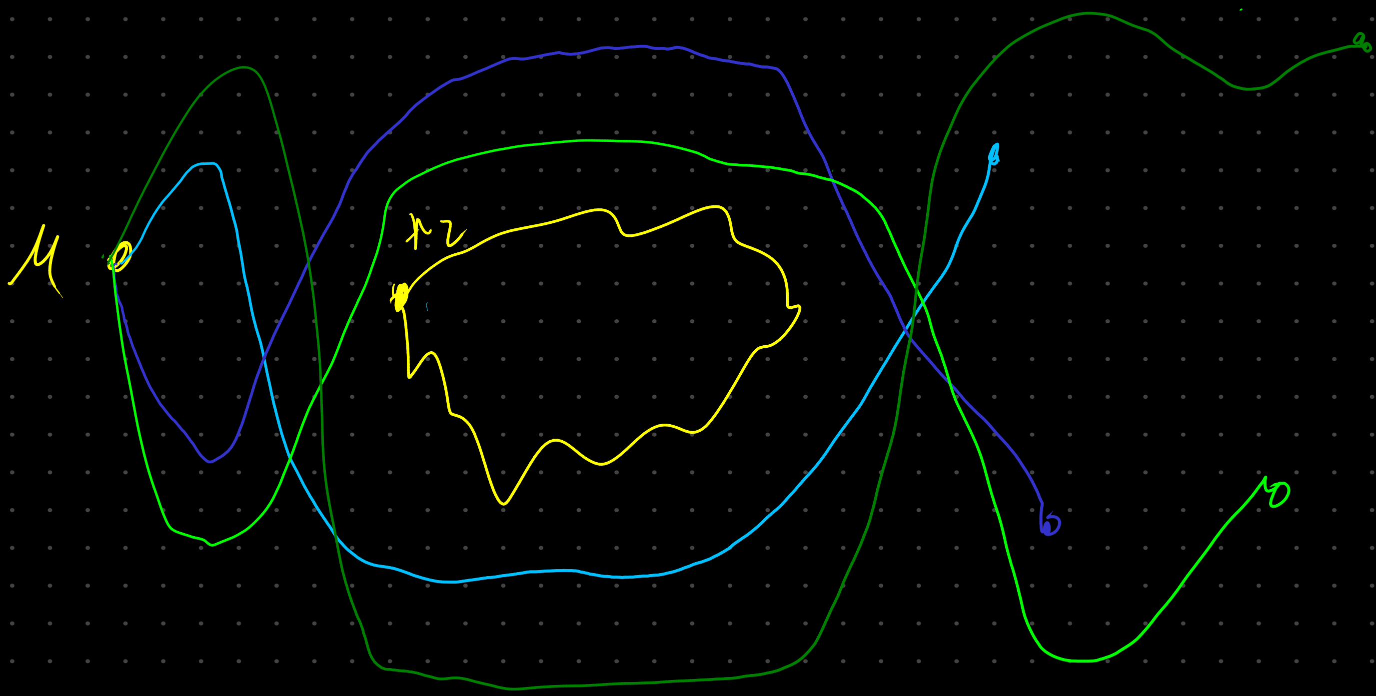
can't prove the left edge ⇒ We can have the not example leading to Lemma 2.3 only edges linking $L(\ell(v)) \neq L(\ell(w))$ $(v,w) \in E(h)$

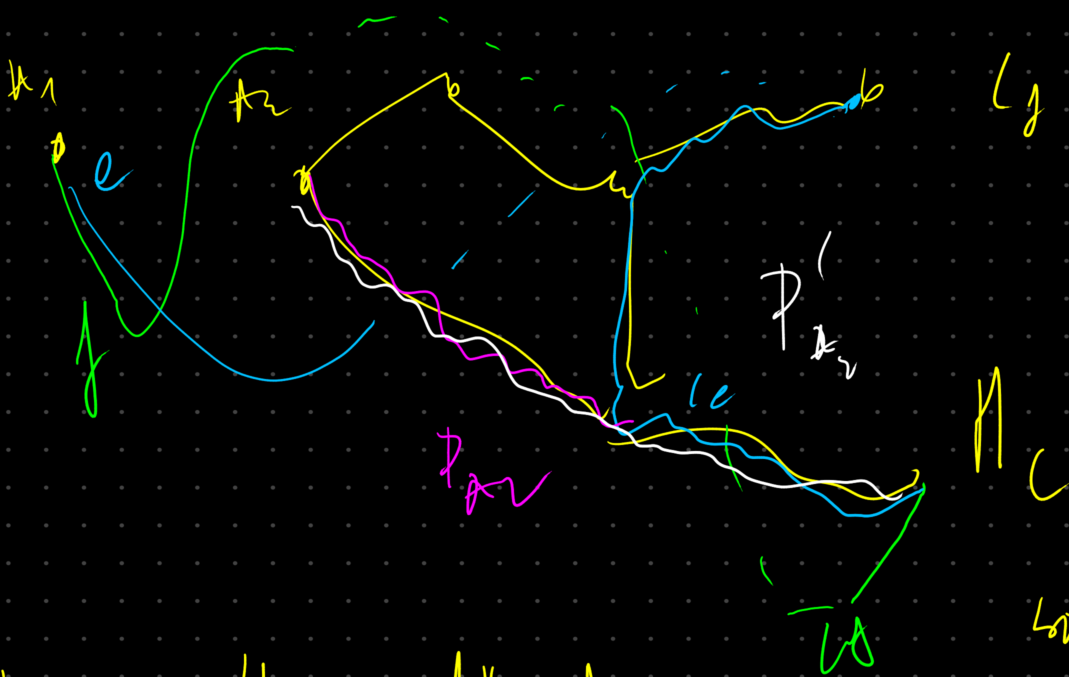




⇒



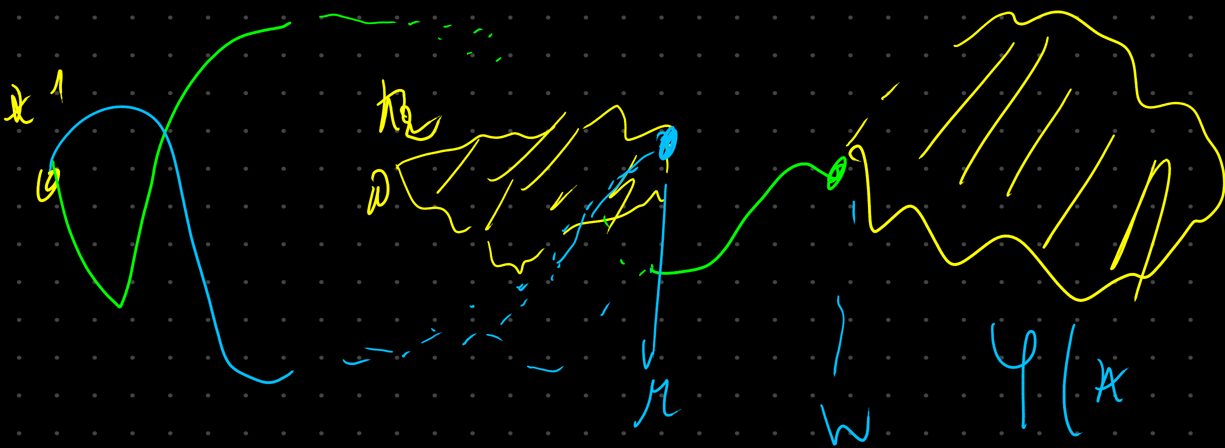


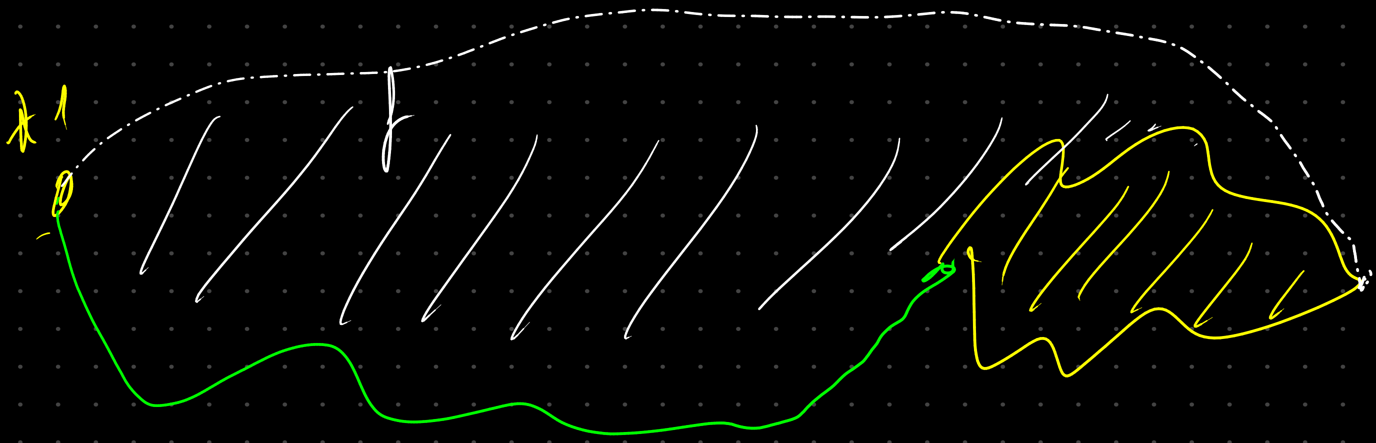


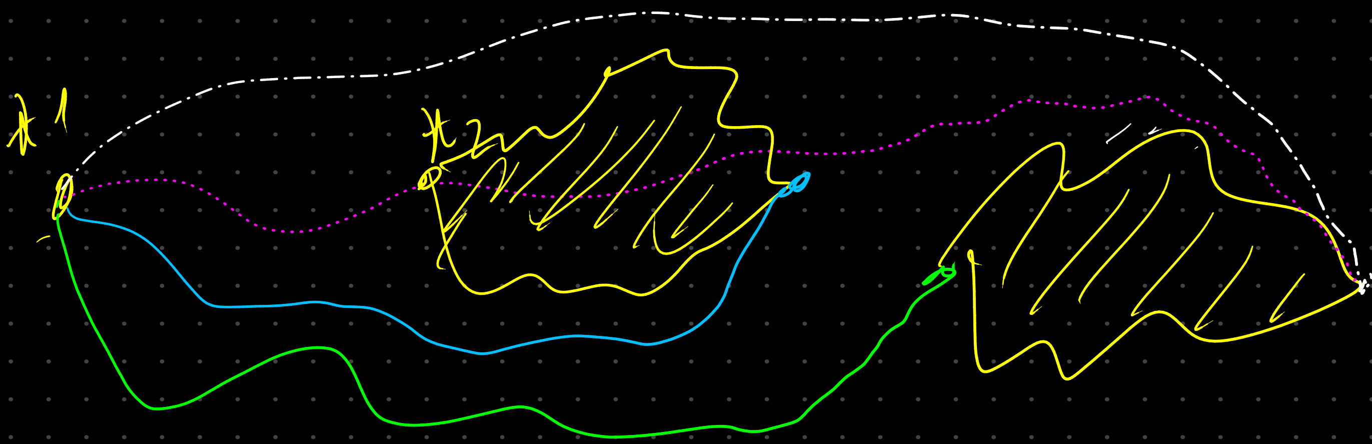
C_j and P'_{A_1} share no edge and C_j above A_2

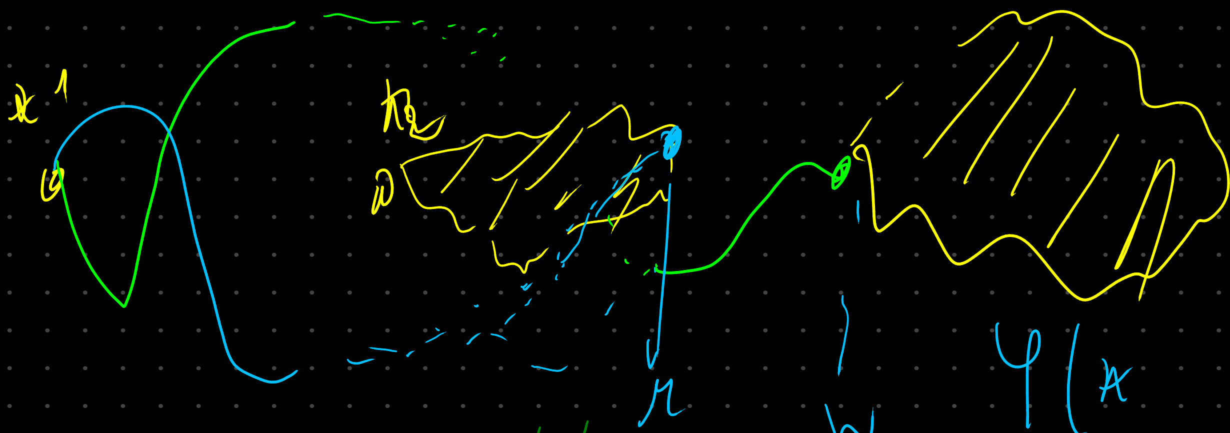
$C_j > P_{A_2}$ but no'

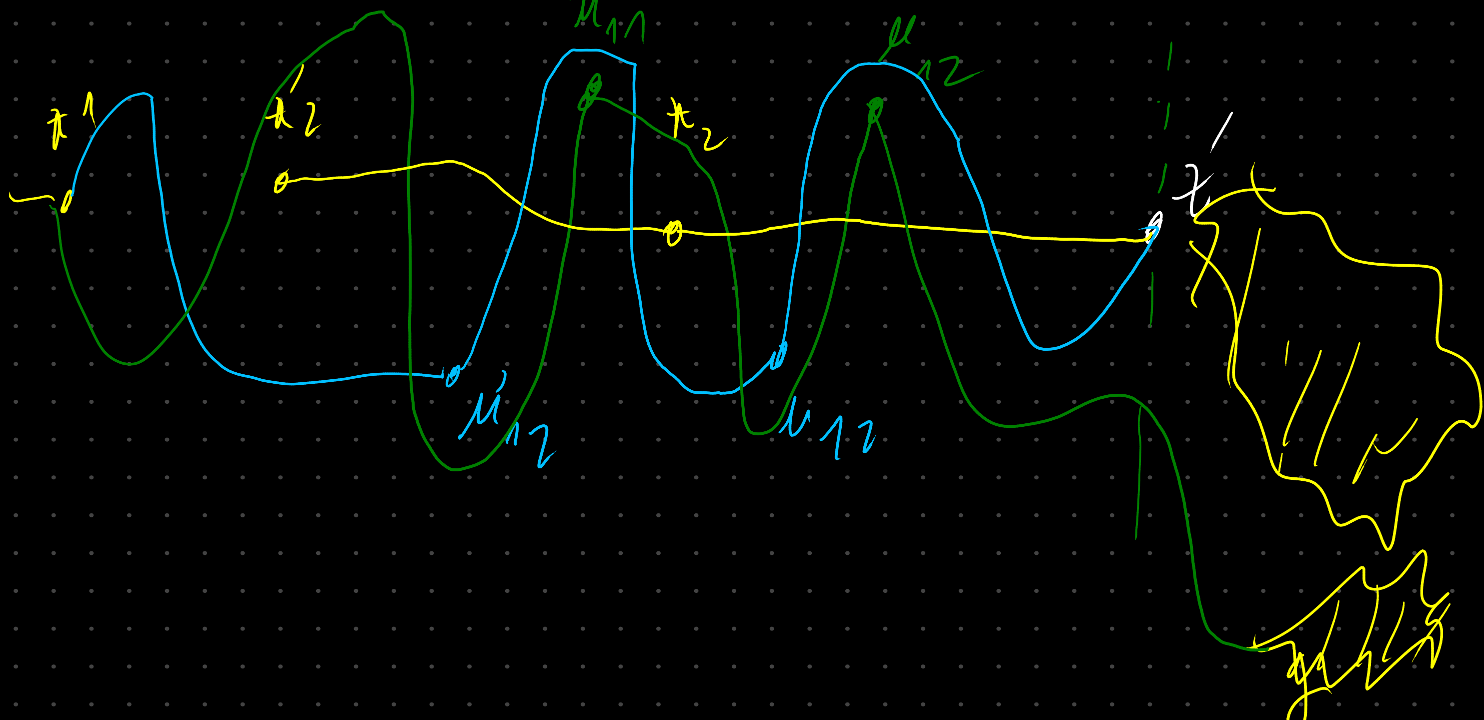
what would be diff for multiple cusps?



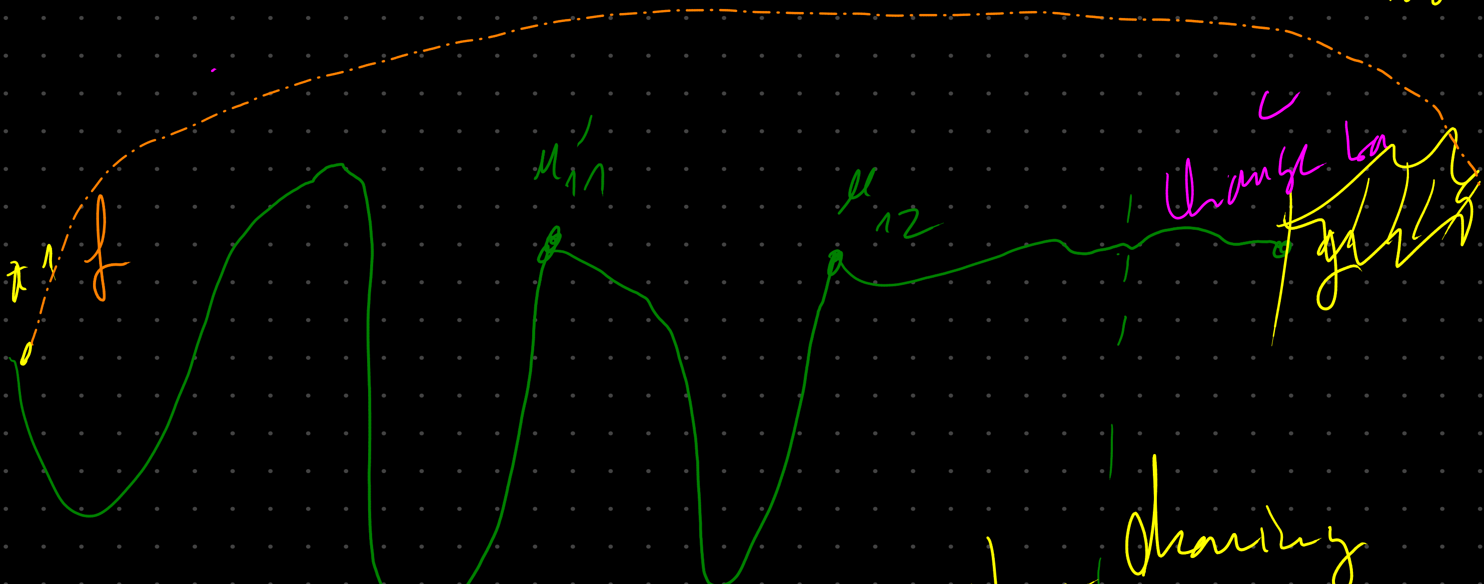

x1
f


x1
x2

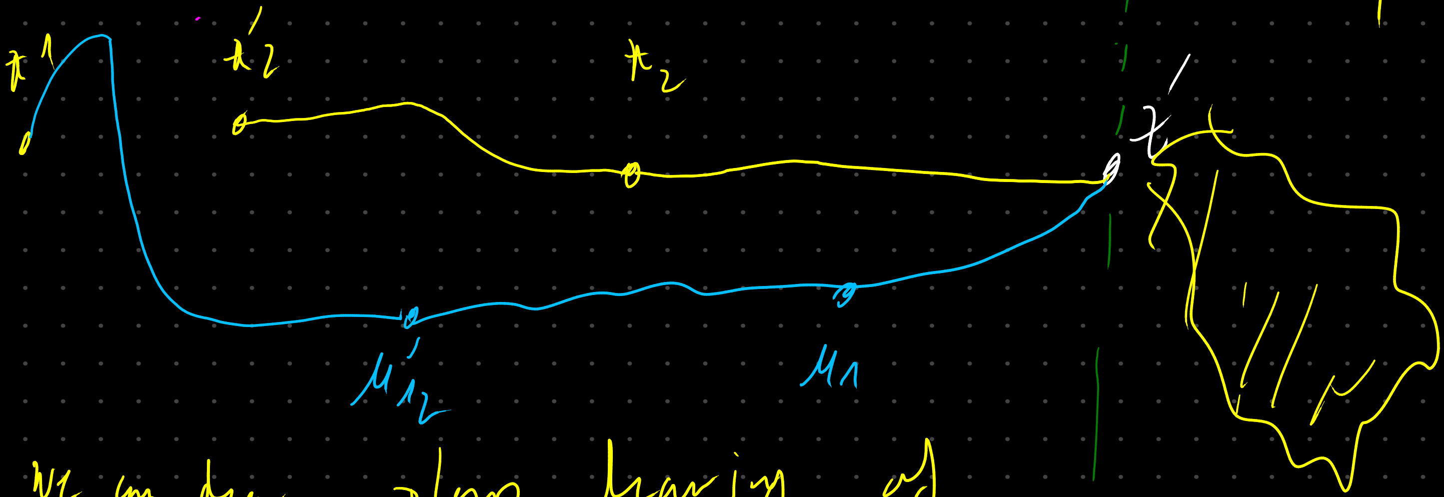

x1
x2
u
w
φ(A


x1
x2
u11
u12
x2
u'12
u12

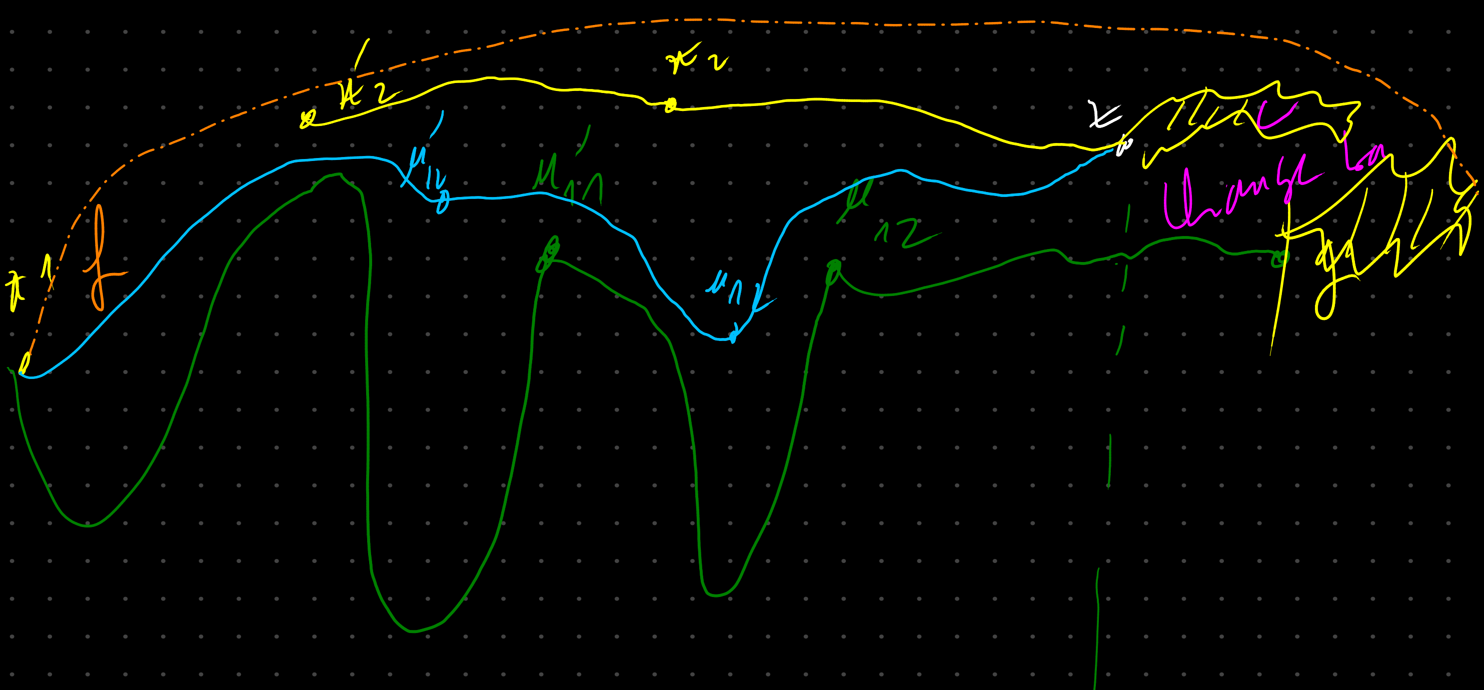
let's assume green H has π-max vertex b/w the two.



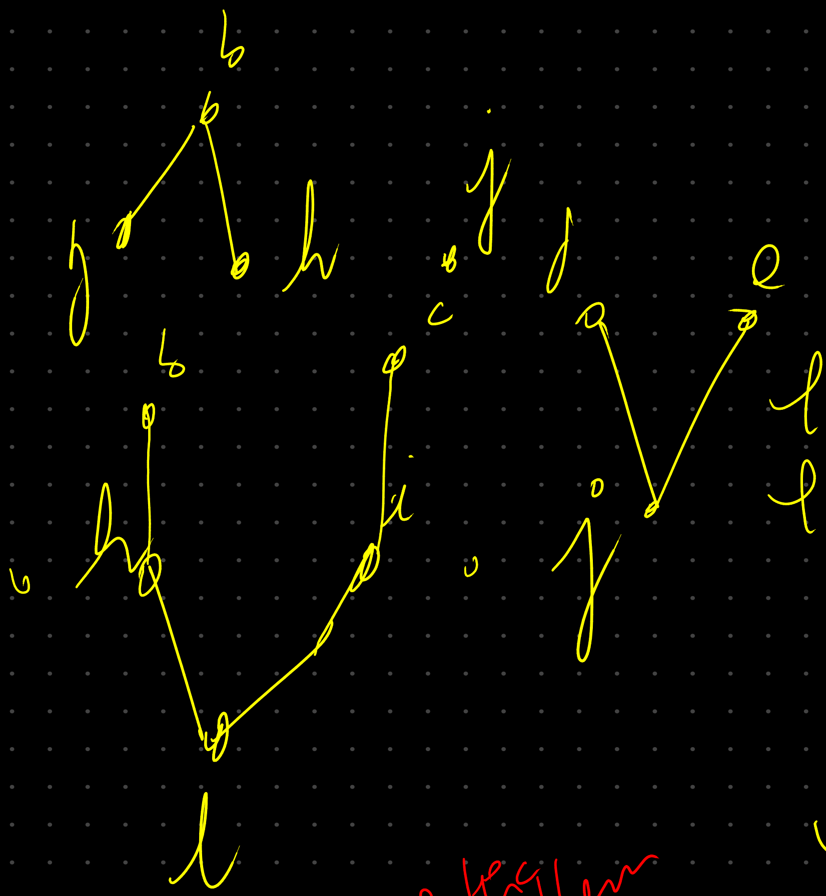
we can draw a planar drawing of green H



we can draw a planar drawing of blue H



intanglement can be removed only with edges $e(u,v)$ with $L(l(u)) \neq L(l(v))$



$\varphi(bd) = \varphi(hj)$

$\varphi(be) = \varphi(hi)$

$\varphi(cd)$ $\quad \varphi(jh) = 1$

$\varphi(ij)$

$\varphi(ji) = 0$

$\varphi(hi) = 1$

$\varphi(bd) = 0 \quad \varphi(cd)$

$\varphi(be) = 0 \quad = \varphi(ce)$

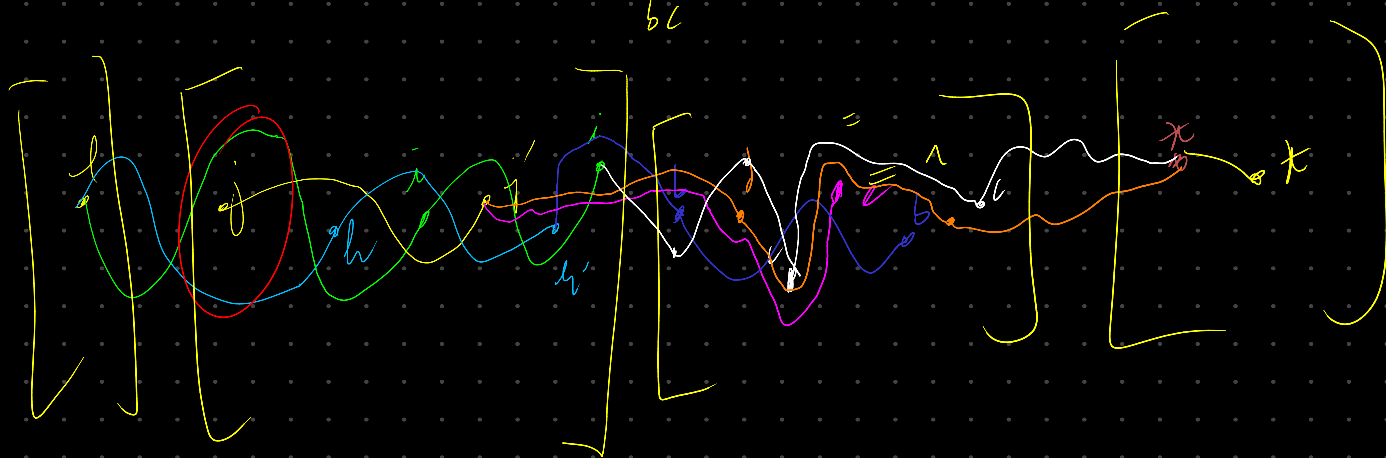
$\varphi(ij)$

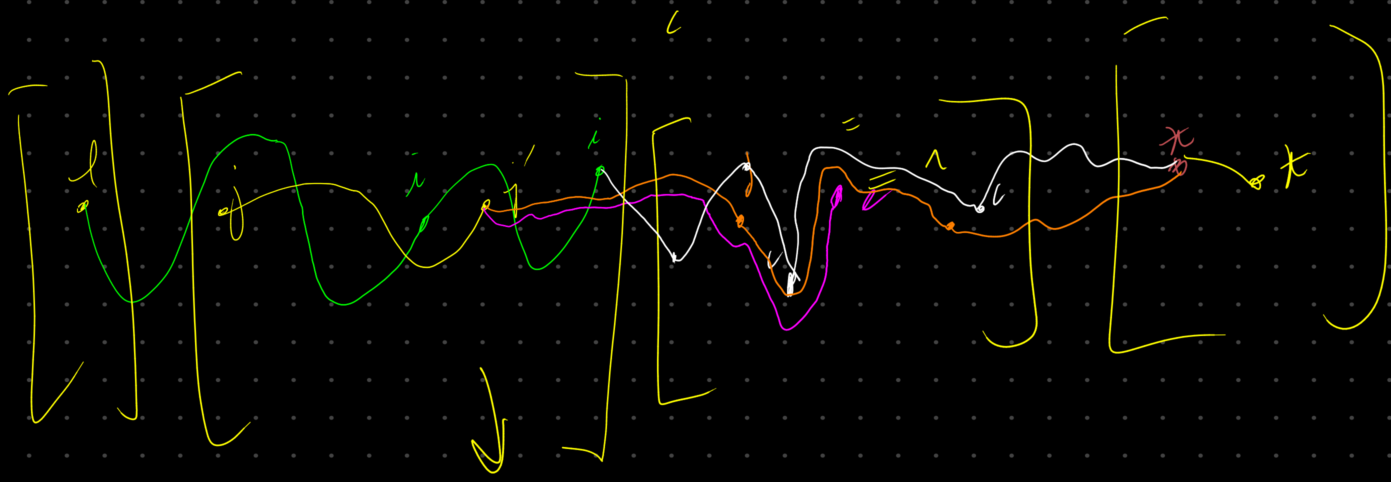
source of the problem

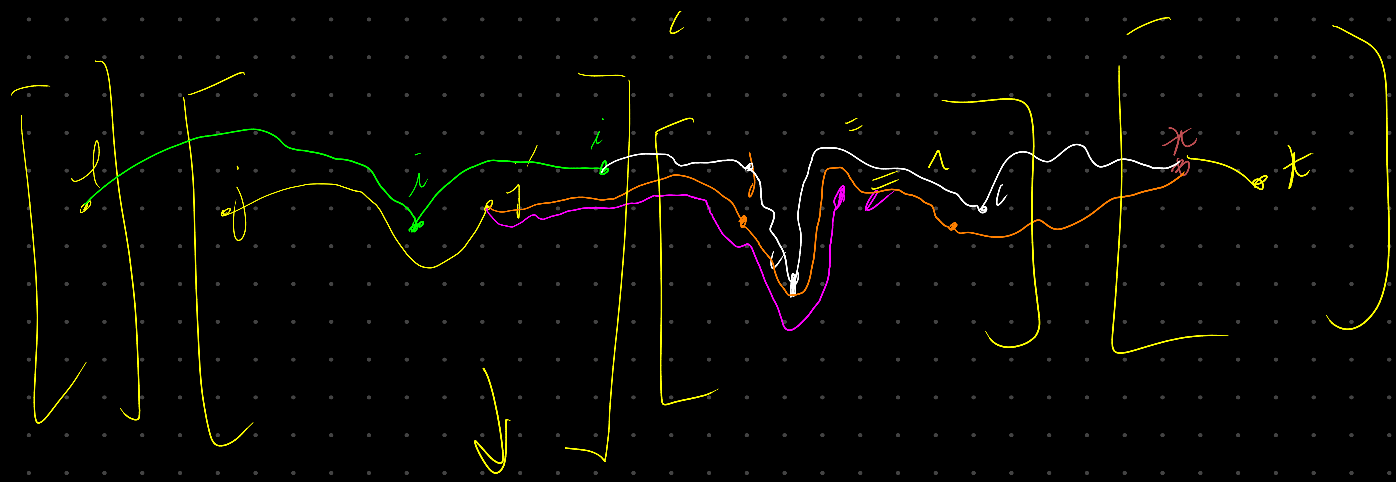
$d = 1$

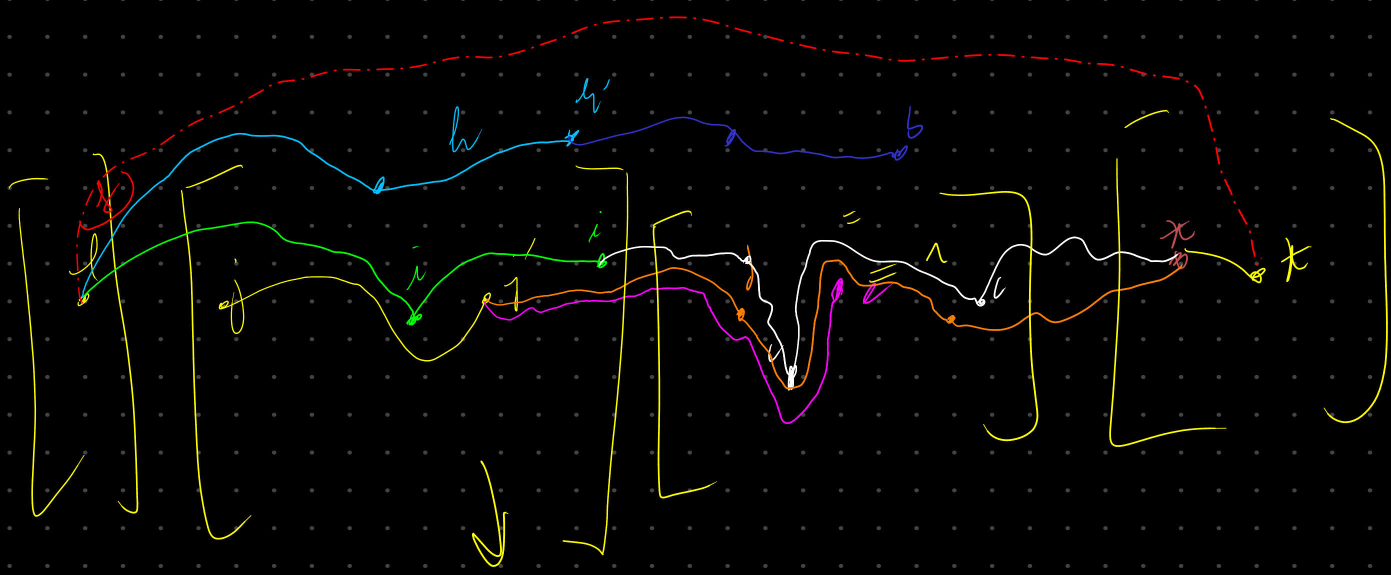
be

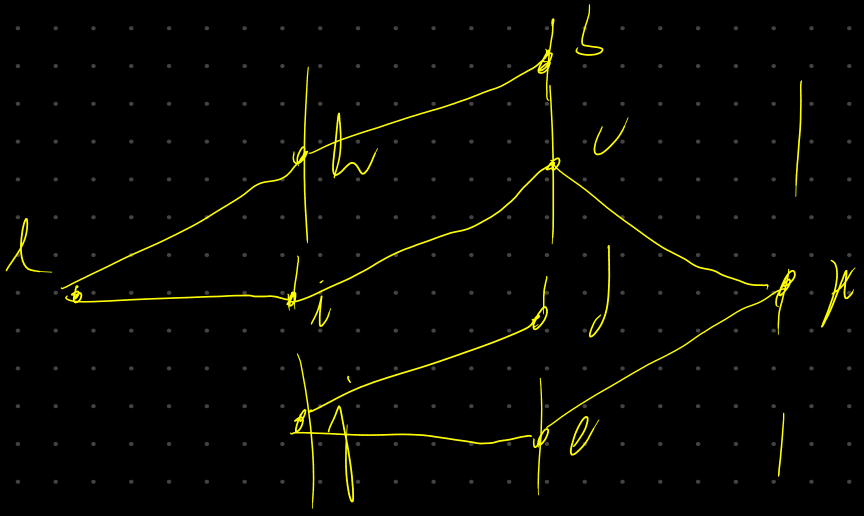
bc

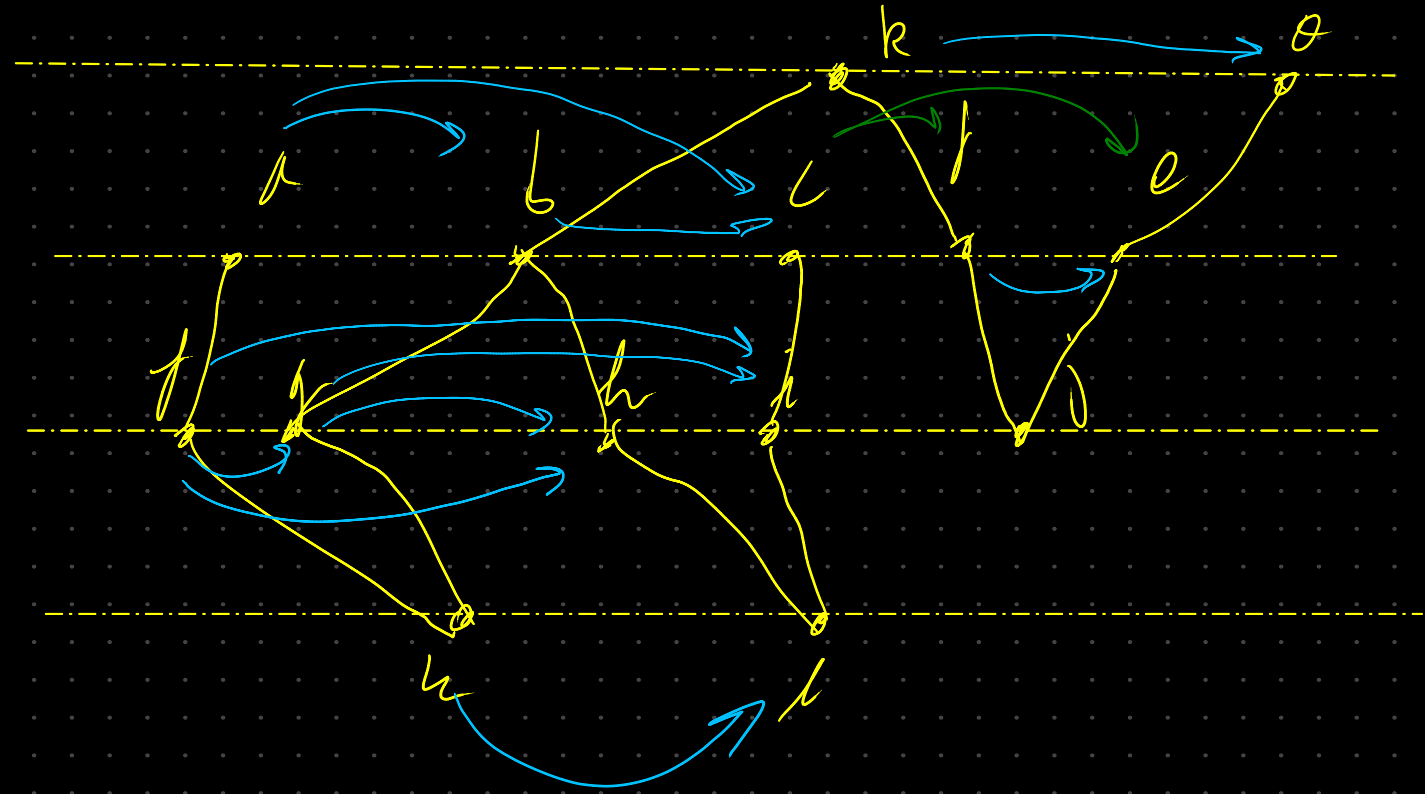


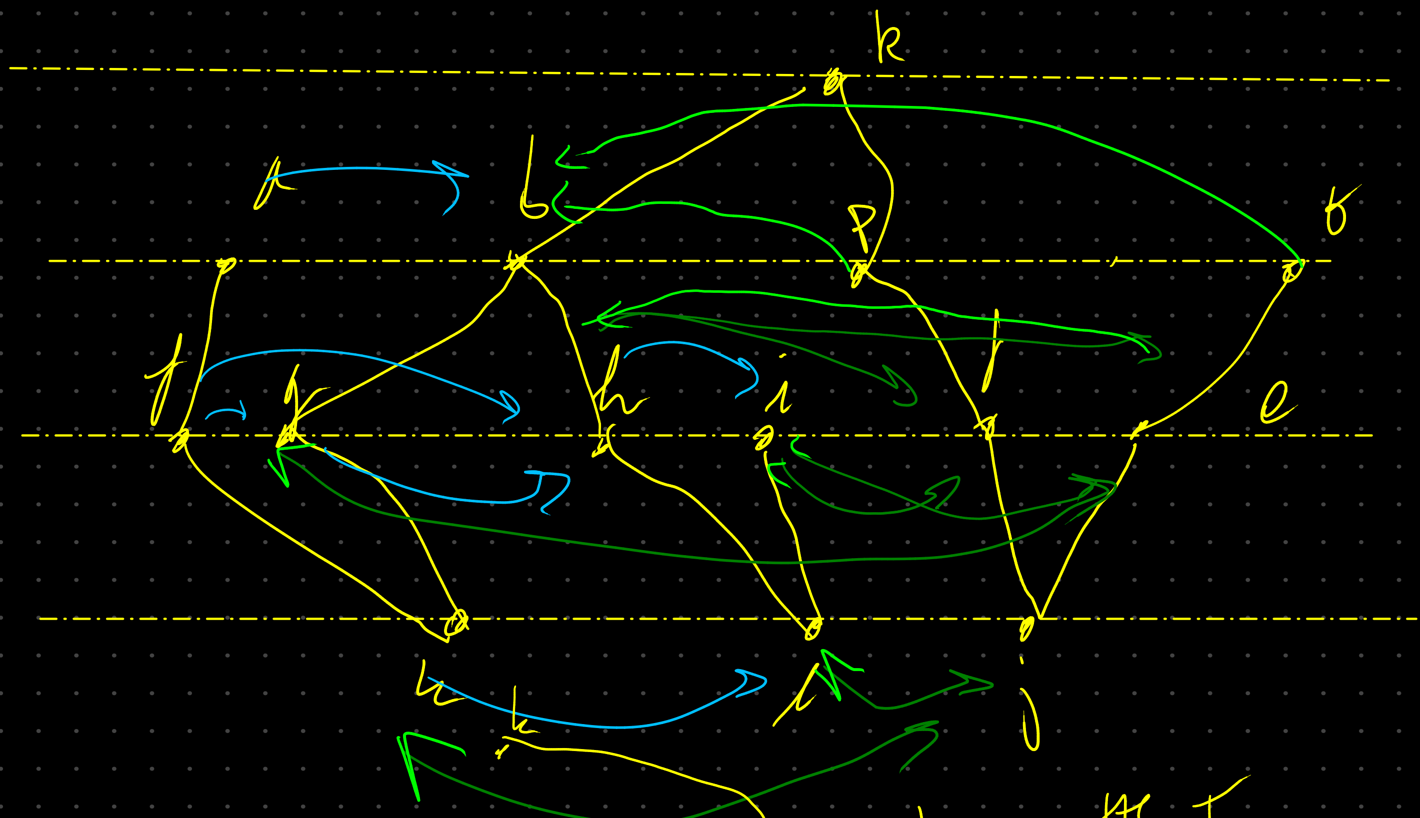












No issue: I have an assumption that having a vertex that is not adjacent to both levels $l-1, l+1$ with $l(v) = l$ and in l there is adjacent edges

$\Rightarrow$ leads to a cyclic order